

To thy happy children of the future, those of the past send greetings.

Alma Mater

At the University of Illinois at Urbana-Champaign, we fundamentally believe that we have a right to make the university better for all future students. That belief is a message etched into our Alma Mater and makes up the DNA of our course staff. As such, we created the coursebook. The coursebook is a free and open systems programming textbook that anyone can read, contribute to, and modify for now and forever. We don't think information should be behind a walled garden, and we truly believe that complex concepts can be explained simply and fully, for anyone to understand. The goal of this book is to teach you the basics and give you some intuition into the complexities of systems programming.

Like any good book, it isn't complete. We still have plenty of examples, ideas, typos, and chapters to work on. If you find any issues, please file an issue or email a list of typos to CS 341 course staff, and we'll be happy to work on it. We are constantly trying to make the book better for students a year and ten years from now.

This work is based on the original SystemProgramming wiki coursebook. All these people's hard work is included in the section below.

Oh and the duck? Keep reading until synchronization :).

Thanks again and happy reading!

– Bhuvy

1.1 Authors

Bhuvan Venkatesh <bhuvan.venkatesh21@gmail.com>

Lawrence Angrave <angrave@illinois.edu>

joebenassi <joebenassi@gmail.com>

jakebailey <zikaeroh@gmail.com>

Ebrahim Byagowi <ebrahim@gnu.org>

Alex Kizer <the.alex.kizer@gmail.com>

dimyr7 <dimyr7.puma@gmail.com>

Ed K <ed.karrels@gmail.com>

ace-n <nassri2@illinois.edu>

josephmilla <jjtmilla@gmail.com>

Thomas Liu <thomasliu02@gmail.com>

Johnny Chang <johnny@johnnychang.com>
goldcase <johnny@johnnychang.com>
vassimladenov <vassi1995@icloud.com>
SurtaiHan <surtai.han@gmail.com>
Brandon Chong <bchong95@users.noreply.github.com>
Ben Kurtovic <ben.kurtovic@gmail.com>
dprorok2 <dprorok2@illinois.edu>
anchal-agrawal <aagrawa4@illinois.edu>
daeyun <daeyunshin@gmail.com>
bchong95 <bschong2@illinois.edu>
rushingseas8 <georgealeks@hotmail.com>
lukspdev <llluuukke@gmail.com>
hilalh <habashi2@illinois.edu>
dimyr7 <dimyr7@hotmail.com>
Azrakal <genxswordsmen@hotmail.com>
G. Carl Evans <gcevans@gmail.com>
Cornel Punga <cornel.punga@gmail.com>
vikasagartha <vikasagartha@gmail.com>
dyarbrough93 <dyarbrough93@yahoo.com>
berwin7996 <berwin7996@gmail.com>
Sudarshan Govindaprasad <SudarshanGp@users.noreply.github.com>
NMyren <ntmyren@gmail.com>
Ankit Gohel <ankitgohel1996@gmail.com>
vha-weh-shh <bhaweshchhetri1@gmail.com>
sasankc <sasank.chundi@gmail.com>
rishabhjain2795 <rishabhjain2795@gmail.com>
nickgarfield <nickgarfield@icloud.com>
by700git <aaabox@yeah.net>
bw-vbnm <bwang19@illinois.edu>
Navneeth Jayendran <jayndrn2@illinois.edu>
Joe Benassi <joebenassi@gmail.com>
Harpreet Singh <hshssingh4@gmail.com>
FenixFeather <thomasliu02@gmail.com>
EntangledLight <bdelapor@illinois.edu>
Bliss Chapman <bliss.chapman@gmail.com>
zikaeroh <zikaeroh@gmail.com>
time bandit <radicalrafi@gmail.com>
paultgibbons <paultgibbons@gmail.com>
kevinwang <kevin@kevinwang.com>
cPolaris <cPolaris@users.noreply.github.com>
Zecheng (張澤成) <zzhan147@illinois.edu>
Wieschie <supernova190@gmail.com>
WeiL <z920631580@gmail.com>
Graham Dyer <gdyer2@illinois.edu>
Arun Prakash Jana <engineerarun@gmail.com>
Ankit Goel <ankitgoel616@gmail.com>
Allen Kleiner <akleiner24@gmail.com>
Abhishek Deep Nigam <adn5327@users.noreply.github.com>
zmmille2 <zmmille2@gmail.com>
sidewallme <sidewallme@gmail.com>
raych05 <raymondcheng05@gmail.com>
mmahes <malinixmahes@gmail.com>

mass <amass1212@gmail.com>
kovaka <jakelagrou@gmail.com>
gmag23 <gmag23@gmail.com>
ejian2 <ejian2@illinois.edu>
cerutii <marc.ceruti@gmail.com>
briantruong777 <briantruong777@gmail.com>
adevar <adevar2@illinois.edu>
Yuxuan Zou (Sean) <yzouac@connect.ust.hk>
Xikun Zhang <xikunz2@illinois.edu>
Vishal Disawar <disawar2@illinois.edu>
Taemin Shin <cprayer@naver.com>
Sujoy Patwardhan <sujoy.patwardhan@gmail.com>
SufeiZ <sufheizhang92@gmail.com>
Sufei Zhang <sufheizhang92@gmail.com>
Steven Shang <sstevenshang@users.noreply.github.com>
Steve Zhu <st.zhu1@gmail.com>
Sibo Wang <sibowsb@gmail.com>
Shane Ryan <shane1027@users.noreply.github.com>
Scott Bigelow <epheph@gmail.com>
Riyad Shauk <riyadshauk@users.noreply.github.com>
Nathan Somers <nsomers2@illinois.edu>
LieutenantChips <vkaraku2@illinois.edu>
Jacob K LaGrou <jakelagrou@gmail.com>
George <ruan3@illinois.edu>
David Levering <dmlevering@gmail.com>
Bernard Lim <bernlm93@users.noreply.github.com>
zwang180 <zshwang0809@gmail.com>
xuanwang91 <LilyBiology2010@gmail.com>
xin-0 <xintong2@illinois.edu>
wchill <wchill1337@gmail.com>
vishnui <vishnui@gmail.com>
tvarun2013 <tvarun2013@gmail.com>
sstevenshang <sstevenshang@users.noreply.github.com>
ssquirrel <lxl_zhang@Hotmail.com>
smeenai <shoaib.meenai@gmail.com>
shrujancheruku <shrujancheruku@gmail.com>
ruiqili2 <ruiqili2@users.noreply.github.com>
rchwlsk2 <rchwlsk2@illinois.edu>
ralphchung <ralphchung2005@gmail.com>
nikioftime <ncwells2@illinois.edu>
mosaic0123 <truffer@live.com>
majiasheng <jjasheng.ma@yahoo.com>
m <cheonghiuwaa@gmail.com>
li820970 <li820970@gmail.com>
kuck1 <kuck1@illinois.edu>
kkgomez2 <kkgomez2@users.noreply.github.com>
jjames34 <James_Jerry1@yahoo.com>
jargals2 <jargals2@illinois.edu>
hzding621 <hzding621@users.noreply.github.com>
hzding621 <hzding621@gmail.com>
hsingh23 <hisingh1@gmail.com>
denisdemaisbr <denis@roo.com.br>

daishengliang <daishengliang@gmail.com>
cucumbur <bomblolism@gmail.com>
codechao999 <brianweis@comcast.net>
chrisshroba <chrisshroba@gmail.com>
cesarcastmore <cesar.cast.more@gmail.com>
briantruong777 <briantruong777@users.noreply.github.com>
botengHY <tengbo1992@gmail.com>
blapalp <pzkmmmh@gmail.com>
bchhetri1 <bhaweshchhetri1@gmail.com>
anadella96 <aisha.nadella@gmail.com>
akleiner2 <akleiner24@gmail.com>
aRatnam12 <ansh.ratnam@gmail.com>
Yash Sharma <yashosharma@gmail.com>
Xiangbin Hu <xhu27@illinois.edu>
WininWin <ezoneid@gmail.com>
William Klock <william.klock@gmail.com>
WenhanZ <marinebluee@hotmail.com>
Vivek Pandya <vivekvpandya@gmail.com>
Vineeth Puli <vpuli98@gmail.com>
Vangelis Tsiatsianas <vangelists@users.noreply.github.com>
Vadim11024 <vadim@mbdsys.com>
Utsav2 <ukshah2@illinois.edu>
Thirumal Venkat <zapstar@users.noreply.github.com>
TheEntangledLight <bdelapor@illinois.edu>
SudarshanGp <SudarshanGp@users.noreply.github.com>
Sudarshan Konge <6025419+sudk1896@users.noreply.github.com>
Slix <slixpk@gmail.com>
Sasank Chundi <sasank.chundi@gmail.com>
SachinRaghunathan <srgnhth2@illinois.edu>
Rémy Léone <remy.leone@gmail.com>
RussellLuo <russelluo@gmail.com>
Roman Vaivod <littlewhywhat@gmail.com>
Rohit Sarathy <rohit@sarathy.org>
Rick Sheahan <bomblolism@gmail.com>
Rakhim Davletkaliyev <freetonik@gmail.com>
Punitvara <punitvara@gmail.com>
Phillip Quy Le <pitlv2109@gmail.com>
Pavle Simonovic <simonov2@illinois.edu>
Paul Hindt <phindt@gmail.com>
Nishant Maniam <nishant.maniam@gmail.com>
Mustafa Altun <gmail@mustafaaltun.com>
Mohammed Sadik P. K <sadiqpkp@gmail.com>
Mingchao Zhang <43462732+mingchao-zhang@users.noreply.github.com>
Michael Vanderwater <vndrwtr2@users.noreply.github.com>
Maxiwell Luo <maxluoXIII@gmail.com>
LunaMystic <suxianghan@outlook.com>
Liam Monahan <liam@liammonahan.com>
Joshua Wertheim <joshwertheim@gmail.com>
John Pham <newhope11134@gmail.com>
Johannes Scheuermann <johscheuer@users.noreply.github.com>
Joey Bloom <15joeybloom@users.noreply.github.com>
Jimmy Zhang <midnight.vivian@gmail.com>

Jeffrey Foster <jmfoste2@illinois.edu>
James Daniel <james-daniel@users.noreply.github.com>
Jake Bailey <zikaeroh@gmail.com>
JACKHAHA363 <luyuchen.paul@gmail.com>
Hydrosis <badda2k@gmail.com>
Hong <plantvsbird@gmail.com>
Grant Wu <grantwu2@gmail.com>
EvanFabry <Evan.Fabry@gmail.com>
EddieVilla <EddieVilla@users.noreply.github.com>
Deepak Nagaraj <n.deepak@gmail.com>
Daniel Meir Doron <danielmeirdoron@gmail.com>
Daniel Le <GreenRecycleBin@gmail.com>
Daniel Jamrozik <djamro2@illinois.edu>
Daniel Carballal <danielenriquecarballal@gmail.com>
Daniel <DTV96Calibre@users.noreply.github.com>
Daeyun Shin <daeyun@daeyunshin.com>
Creyslz <creyslz@gmail.com>
Christian Cygnus <gamer00@att.net>
CharlieMartell <charliecmartell@gmail.com>
Caleb Bassi <calejbassi@gmail.com>
Brian Kurek <brkurek@gmail.com>
Brendan Wilson <brendan.x.wilson@gmail.com>
Bo Liu <boliu1@illinois.edu>
Ayush Ranjan <ayushr2@illinois.edu>
Atul kumar Agrawal <ms.atul1303@gmail.com>
Artur Sak <artursak1981@gmail.com>
Ankush Agarwal <ankushagarwal@users.noreply.github.com>
Angelino <angelino_m@outlook.com>
Andrey Zaytsev <andzaytsev@gmail.com>
Alex Yang <alyx.yang@gmail.com>
Alex Cusack <cusackalex@gmail.com>
Aidan Epstein <aidan@jmad.org>
Ace Nassri <ace.nassri@gmail.com>
Abdullahi Abdalla <abdalla6@illinois.edu>
Aneesh Durg <durg2@illinois.edu>
Assassin Eclipse <hungwoei96@hotmail.com>
Eric Cao <eric7252000@gmail.com>
Raphael Long <rafilong42@gmail.com>
williamsentosa95 <38774380+williamsentosa95@users.noreply.github.com>
Pradyumna Shome <pradyumna.shome@gmail.com>
Benjamin West Pollak <benjaminwpollak@gmail.com>
姜凡越 Pengyue Jiang <pengyue3@illinois.edu>
Andrew Orals <aorals2@illinois.edu>
Elijah Mock <emock3@illinois.edu>
Cay Zhang <13341339+Cay-Zhang@users.noreply.github.com>
Wu Shuwen <mikewushuwen@outlook.com>
Leng Ang Tan <2025495tan@aupp.edu.kh>